

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 - 2019 (Even Semester)

Innovative Practices Description

Degree, Semester& Branch: VIII Semester B.E. ECE B

Course Code & Title: EC6802 Wireless Networks

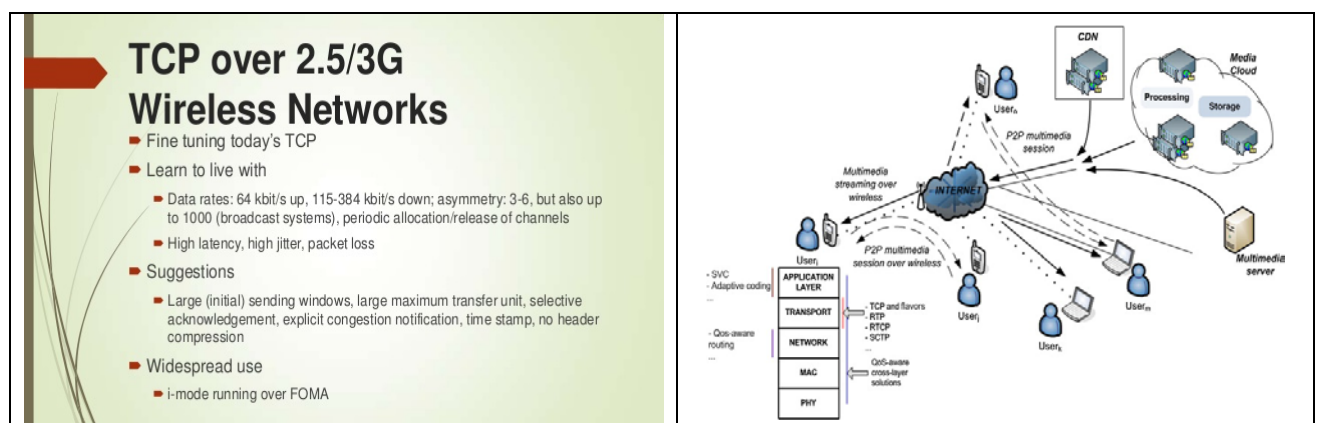
Name of the Faculty member: Mr.V.Rajesh

Name of the Topic: TCP over 3G wireless networks

Name of the Innovative Practice: NPTEL video

Description:

- TCP over 2.5G and 3G wireless networks. Abstract: Transmission control protocol (TCP) by IETF is one of the most widely used transport layer protocols in Internet. Much development and deployment activity has centered around GPRS, UMTS and IMT-2000, also referred to 2.5G/3G wireless networks.
- Transmission control protocol (TCP) by IETF is one of the most widely used transport layer protocols in Internet. Much development and deployment activity has centered around GPRS, UMTS and IMT-2000, also referred to 2.5G/3G wireless networks. However, TCP has not been designed bearing in mind wireless networks.
- Especially, flow control features can perform less than optimally over wireless interfaces. A number of TCP optimization techniques have been studied to enhance the TCP performance for various wireless environments. This paper proposes a profile of such techniques, particularly effective for to 2.5G/3G wireless networks.
- Wireless link losses result in poor TCP throughput since losses are perceived as congestion by TCP, resulting in source throttling. In order to mitigate this effect, 3G wireless link designers have augmented their system with extensive local retransmission mechanisms.
- In addition, in order to increase throughput, intelligent channel state based scheduling have also been introduced. While these mechanisms have reduced the impact of losses on TCP throughput and improved the channel utilization, these gains have come at the expense of increased delay and rate variability.
- We comprehensively evaluate the impact of variable rate and variable delay on long-lived TCP performance. We propose a model to explain and predict TCP's throughput over a link with variable rate and/or delay. We also propose a network-based solution called Ack Regulator that mitigates the effect of variable rate and/or delay without significantly increasing the round trip time, while improving TCP performance by up to 100%.



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	1	1	-	-	1	-	-	-	1	1	-	1

CO3: The students will be able to Describe about various techniques in Mobile Transport Layers.

References:

1. https://www.youtube.com/watch?v=XPCJzmhhgKg&index=65&list=PLKIo5324t_alfdqOdPcXwpNAF9e8Og6hV
2. <https://www.youtube.com/watch?v=FUgdF-2V8Cw>
3. <https://www.youtube.com/watch?v=1I3MEqlVqMc>